

D6 - Generalization, God model, limits

Season-to-season: the decision-relevant OOD electricity is strongly seasonal; which corridors bind changes with the season
Jan-May 87.8, Apr-Jul 92.7, Jun-Aug 62.5
calibration stable, error winner flips

Year-to-year: stress limit, not the operating regime
calendar 86.9 vs 80.6, full 69.7 vs 58.6
8,978 params → frequently train-once
Generalize-a-year is the wrong question

3 pairs = 1 primary + 2 probes
primary HLUBAVG to HL_PAK: full comparison, 5 seeds
probes NORTH 86.6%, WEST 91.0%; proposed model only claim = calibration consistency, not a ranked comparison

Sensitivity lag sets 24 / 24-48 / 24-48-168 lead 1, 2, 4, 12 hours → coverage stable → 89.5%

God model: fuse every model's winning axis
time in query, identity-only, full fusion, train on Winkler, adaptive conformal, factorization, normalized conformal, min-width routing, penalty tuning
NONE beats SPARC at 5 seeds → SPARC is at the optimum

Conformal defense – the result CQR-LQR reaches only 79.3% (up from <73%)
WHAT: post-hoc patch – hold out a calibration set, shift the band to hit target coverage, model-agnostic
OBJECTION: 'just wrap the linear model and it hits 90% too'
TEST: wrap LQR in split conformal on the SAME test half

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Protocol CPU, chronological 70-15-15 5 seeds, pure-pinball AQL ablation = remove one part, retain, measure the drop

Results SPARC coverage 87.9%, Winkler 25.72, AQL 1.335, AQCR 0.507

Findings: best calibrated, beats linear AND deep AQL parity with LQR, beats all deep nets best coherence; 7 to 9 times smaller significant: coverage 16.50, p 0.003

Honest negatives LQR edges AQL slightly; CPS parity scope: central 90% band, extreme events out of scope

Ablations No attention: coverage 87.9 to 78.6, AQL flat = attention is the load-bearing calibration mechanism No identity <7.55pp vs no-identity -0.97pp = WHICH blinds beats HOW large No lags equals full = SPARC is at the optimum

Mechanism attention 0.23 to 0.26 on top-3 μ slots the uniform 0.020

Why it matters for trading a band claiming 90% but holding 73% leaves 27% of hours under-hedged low mean error does NOT rescue bad intervals

THE METRIC-PARADOX AQL is middle-dominated → LQR wins it Winkler taxes the band edges → SPARC wins it the reported winner depends on the metric

Empirical interval coverage Coverage = (1/N) * Σ I(L ≤ y ≤ U) For a 90% interval: L = Q_{0.05} and U = Q_{0.95} 88 of 100 hours inside = 88%

Winkler score for a 90% interval W = U - L, If L ≤ y ≤ U W = U - L + 20(L - y), if y < L W = U - L + 20(y - U), if y > U

Adjacent quantile crossing rate AQCR = crossings / adjacent quantile pairs A crossing occurs when q₁ > q₂

AQL / pinball pri(y - q) = (y - q) / (24-48-168 lead 1, 2, 4, 12 hours → coverage stable → 89.5%)

Why attention, not something else?

MLP over the constraint features REJECTED: no explicit weight per constraint cannot tell you which line drove the forecast

IBN or Transformer over the constraint set REJECTED: a set has no order; first and last are interchangeable imposes an order that does not exist

ATTENTION – CHOSEN handles a variable-size set gives a per-path weight to each constraint order-free (permutation-invariant) the weights play the role of learned ASF → interpretable by construction

Inputs top-50 constraint slots (μ, identity, kv, flow) temporal, Fourier, 3 lagged spreads, path embedding

Shadow-price pathway per-slot keys and values the VALUES carry the shadow-price (μ) signal

Identity pathway MEAN-POOLED over binding constraints fixed size because K varies hourly captures WHICH set of corridors binds

D3 - The model

Query = the path asking which constraints matter for me

Temporal encoder

Lagged-spread encoder

Quantile head soft flat head + crossing penalty AQCR 0.507 = LOWEST of all learned models

Training loss $\mathcal{L} = \mathbb{E}_t \mathbb{E}_p \text{pr}(y_t - \hat{y}_t) + \lambda \cdot \text{crossing penalty}$
 $\text{pr}(u) = u \cdot (1 - \mathbb{I}[u < 0])$
penalty is training-only, never a metric

Anchor hard head tested on SPARC Winkler 17.645 vs soft 17.524 → kept soft

Weighted blend of the value signals = $\sum_i \alpha_i \cdot \mu_i$ same form as $\sum_i \Delta SF_i \cdot \mu_i$

Raw concatenation of the value signals REJECTED: the number of binding constraints changes every hour the network needs a fixed-size input

Mean-pool everything REJECTED: treats every binding line as equally important for every path destroys selectivity; a line near the sink should matter more than one far away

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Anchor: Yu et al., Austria, MEDIAN

Known piecewise settlement formula embedded as differentiable layers RULE-EMBEDDINGS

Hard hierarchical head non-crossing by construction, AQCR = 0

Never reports coverage, Winkler, CRPS, PIT

What is given: the lagged μ (a published output) What is learned: ASF as attention weights

SPARC is constraint-aware NOT market-rule-informed

LMP identity is EXACT at the optimum LMP_i = lambda + sum_k SF_k μ_k μ_k, lambda, SF_k μ_k are the primal and dual quantities of the optimization

But they are ENDOGENOUS defined only by solving the clearing no closed-form function from raw inputs

Rule is real, but NOT translatable so LEARN the exposure weights

Our move: condition on the PAST published clearing outputs most recent snapshot strictly before the target hour's own μ is NOT used – it exists only after that hour clears, so using it would leak

Delta-SF is not published for each source-destination pair depends on topology + operating point

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Path spread ΔLMP_i = d LMP_i - LMP_s = $\sum_k (SF_k \cdot d - SF_k \cdot s) \cdot \mu_k$ = $\sum_k \Delta SF_k \cdot \mu_k$

Differential shift factor ΔSF_k = SF_{k,d} - SF_{k,s} Example: 0.6 - 0.1 = 0.5 If μ_k = \$20/MWh, contribution = \$10/MWh

A cancels exactly spread = pure congestion differential

Why THIS IS THE RIGHT ASSUMPTION μ is the market's own signal of where and how badly the grid is constrained

Outcome tick price-only models: good center, bad risk LQR coverage 73.2%, crosses SPARC 87.9%, crosses 0.507%

Example: road at capacity one more MW costs +\$20 therefore μ = \$20/MWh

Selection: per hour, per constraint sort by its own μ descending, take top 50 binding only, snapshot lagged one hour

Needs domain understanding, not just sending prices to ML. And that is shown by our results

ERCOT day-ahead, nodal

Nodal price LMP_i = λ + $\sum_k SF_k \cdot \mu_k$

Congestion component $\sum_k SF_k \cdot \mu_k$ location-specific

Shadow price μ_k = marginal cost of relaxing constraint k by 1 MW

Energy component λ same at every node

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S2 - Shadow prices (μ) drive location/prices which constraints bind, and their μ_k are the mechanism → model the market

S3 - The spread price-only μ_k = ΔSF_k - μ_s

S4 - SCED: no formula to embed λ, SF_k, μ_k are optimization outputs

S5 - Anchor (Austria) + our move they embed a rule; we do output-conditioning

S6 - The model constraints → attention → quantiles soft head, least crossings

D2 - SCED + anchor + our move

S7 - Metrics coverage, Winkler, AQL, AQCR the metric-paradox

D4 - Metrics

S8 - Results and mechanism best calibration; attention is load-bearing

S9 - Generalization 3 pairs, seasons, conformal

S10 - God model 10 fusion attempts, all fail

S11 - Decision and limits calibration: first, honest scope

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S0 - Definition SPARC: constraint-aware probabilistic forecaster of ERCOT day-ahead spreads

S1 - How prices form LMP = energy + congestion prices differ by location

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